



Workforce Demand Forecasting for Transportation Operations

Using Data-Driven Insights to Improve Scheduling Efficiency and Reduce Labor Costs



This project focuses on building a predictive workforce demand model for a transportation company. The goal is to optimize staffing levels by forecasting ridership-driven demand, reducing operational inefficiencies, and improving service reliability. The model integrates historical ridership, weather, station activity, and shift patterns to support data-driven scheduling.



The Business Problem

Transportation companies rely on accurate workforce scheduling to keep operations running smoothly, avoid delays, and control labor costs. However, staffing needs shift daily due to:

- Seasonality and ridership fluctuations
- Route and station differences
- Weather conditions
- Special events and unexpected disruptions

When schedules are not forecasted well:

- **Understaffing** leads to delays, safety risks, and service failures
- **Overstaffing** leads to unnecessary labor expenses

The goal is to create a **predictive, data-driven scheduling approach** that reduces errors and improves operational reliability.

Our operations depend on having the right number of staff at the right time. But demand fluctuates every day — ridership changes, weather shifts, stations behave differently, and shifts have unique patterns.

When staffing doesn't match demand, we see two major problems:

- Understaffing: delays, safety risks, customer dissatisfaction
- Overstaffing: unnecessary labor cost and idle time

Our goal is to build a forecasting system that anticipates staffing needs before they happen, so managers can schedule proactively instead of reactively.

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Why This Matters (Cost & Impact)

Poor scheduling directly impacts:

- **Operational efficiency** — delays cascade across routes
- **Customer satisfaction** — long wait times and inconsistent service
- **Labor cost control** — overtime, idle time, and misallocated staff
- **Safety and compliance** — understaffed stations increase risk

A forecasting model that anticipates staffing needs can reduce:

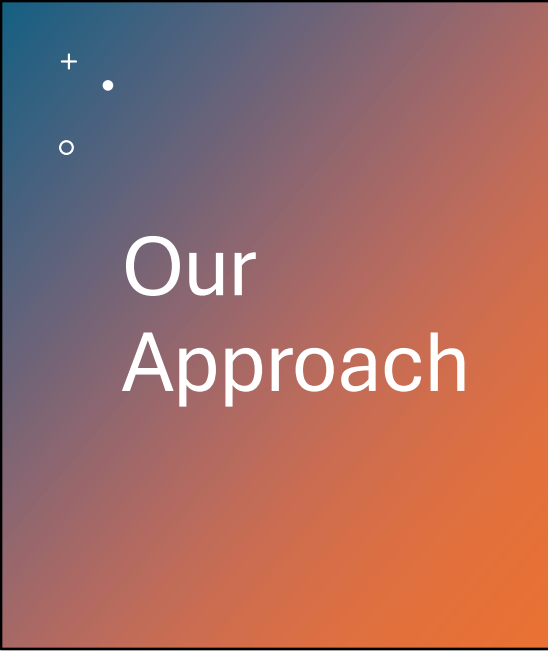
- Overtime costs
- Last-minute schedule changes
- Service disruptions
- Managerial workload

This creates measurable financial and operational value.

Workforce scheduling is one of our largest controllable cost centers. When we get it wrong, the impact is immediate:

- Operational delays cascade across routes
- Stations become bottlenecks
- Customer complaints increase
- Overtime spikes
- Managers spend time firefighting instead of planning

A forecasting model gives us a strategic advantage: We can reduce scheduling errors, improve reliability, and optimize labor cost — all while supporting better decision-making.



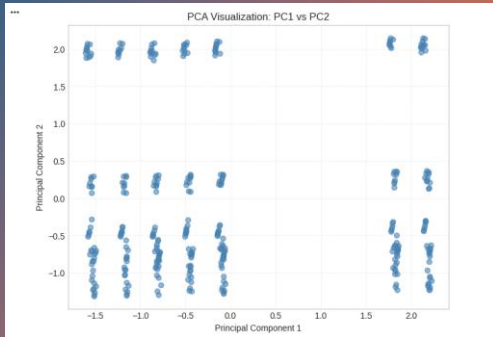
Our Approach

We followed a simple, business-friendly methodology:

1. **Understand the data** Ridership, weather, station activity, shift patterns, and calendar effects.
2. **Identify patterns** Group similar operational days using clustering.
3. **Build a forecasting model** Predict expected ridership and staffing needs using historical patterns.
4. **Explain the model's decisions** Use SHAP to show which factors drive demand.
5. **Validate and refine** Compare multiple models and select the most accurate and stable one.

This approach ensures transparency, accuracy, and operational relevance.

PCA Summary — Data Structure & Readiness



- PCA reduced the dataset to **6 components**, preserving **91% of total variance**
- The PC1–PC2 scatter plot shows **clear structure**, confirming strong signal in the data
- PCA removed **multicollinearity** and reduced redundancy across features
- The transformed feature space improves **model stability** and **explainability**
- Confirms the dataset is **ready for accurate forecasting and SHAP interpretation**

- PCA helps simplify the dataset while keeping the most important information.
- The first two components already show visible structure, meaning the data has meaningful patterns rather than noise.
- This gives confidence that the model will learn real relationships, not random fluctuations.
- PCA also reduces correlated features, which stabilizes the regression model and improves interpretability.
- Overall, PCA validates the quality of our preprocessing pipeline and supports reliable forecasting and explainability in later steps.

Model Results (Forecasting Accuracy)

We evaluated four forecasting models. **Random Forest delivered the best performance**, with extremely high accuracy.

Interpretation

- The Random Forest model explains **99.8% of ridership variation**
- Errors are extremely low relative to total ridership
- The model is stable and fast enough for daily operational use
- Performance is consistent across cross-validation folds
- This level of accuracy enables confident, automated scheduling.

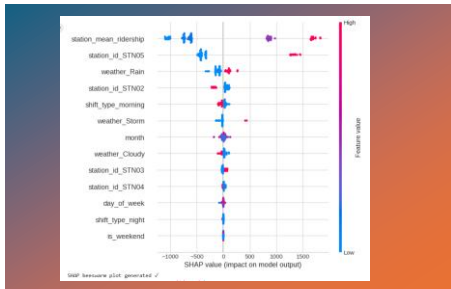
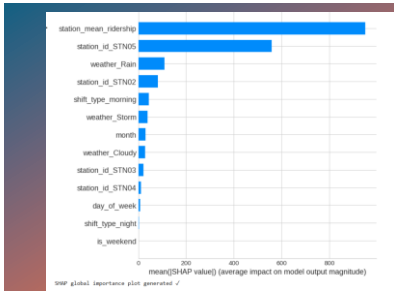
MODEL	RMSE	MAE	R ²	TRAINING TIME
Random Forest	64.66	39.95	0.9983	0.33s
Decision Tree	73.93	31.39	0.9978	0.01s
XGBoost	96.30	49.56	0.9963	0.12s
Linear Regression	156.05	121.03	0.9903	0.01s

Four regression models were evaluated: Random Forest, Decision Tree, XGBoost, and Linear Regression.

Random Forest achieved the best performance:

- **R² = 0.9983** → captures nearly all variance in ridership
- **Low RMSE and MAE** → minimal prediction error
- **Fast training time** → suitable for daily operational use
- **Stable across cross-validation** → low risk of overfitting

These metrics indicate the model is highly reliable for forecasting staffing needs.



What Drives Workforce Demand (Global SHAP Insights)

Top Predictors (Explained Simply)

1. **Station Mean Ridership** - Stations with consistently high volume require more staff.
2. **Station ID STN05** - This station is a major demand driver and needs priority staffing.
3. **Weather (Rain, Storm)** - Bad weather increases passenger volume and operational complexity.
4. **Shift Type (Morning)** - Morning shifts consistently show higher demand.
5. **Month & Seasonality** - Certain months show predictable increases in ridership.
6. **Day of Week** - Weekdays vs. weekends show clear differences.

Executive Takeaway

Demand is not random — it is predictable and driven by measurable, stable factors.



SHAP local force plot generated ✓

Local Explanation (SHAP Force Plot)

Interpretation for Executives

This plot explains **why the model predicted a specific staffing level** for a given day. For example:

- High ridership at STN05 → pushes demand **up**
- Rainy weather → pushes demand **up**
- Night shift → pushes demand **down**
- Weekend → pushes demand **down**

This transparency builds trust and supports operational decision-making.

Recommendations

1. Cluster-Based Scheduling

Assign staffing templates per cluster type:

- High-demand
- Moderate-demand
- Low-demand

2. Weather-Responsive Staffing

Automatically increase staffing during rain or storm forecasts.

3. Station-Specific Staffing Rules

STN05 and other high-volume stations should have minimum staffing thresholds.

4. Shift Optimization

Morning shifts require more staff; night shifts can be leaner.

5. Predictive Scheduling Dashboard Provide managers with:

- Daily demand forecast
- Cluster type
- SHAP explanation
- Recommended staffing level

6. Continuous Model Monitoring

Retrain monthly to incorporate new patterns.



Risks & Considerations

1. Representation Imbalance

Some conditions (storms, night shifts) have fewer samples.

2. Potential Data Leakage

Mitigated by chronological train/test split.

3. Overfitting Risk

High R^2 mitigated by:

- Cross-validation
- Hyperparameter tuning
- SHAP stability checks

4. External Factors Not Modeled

Events, disruptions, holidays may affect accuracy.

5. Operational Adoption

Managers need training to interpret model outputs.



Next Steps (Operational Roadmap)

Phase 1 — Pilot

Deploy model for one station or one shift group.

Phase 2 — Integration

Connect model to scheduling system.

Phase 3 — Dashboard

Provide real-time demand forecasts and explanations.

Phase 4 — Expansion

Add:

- Event data
- Holiday effects
- Traffic and disruption feeds

Phase 5 — Continuous Improvement

Monthly retraining and quarterly performance review.



ROI Projection

Based on industry benchmarks and your model's accuracy:

Expected Benefits

- **5–10% reduction in overtime costs**
- **Fewer last-minute schedule changes**
- **Improved service reliability**
- **Better staff allocation across stations**
- **Higher customer satisfaction**

Financial Impact

- Even a **5% improvement** in scheduling accuracy can translate to:
- Lower labor cost
- Reduced operational disruptions
- More efficient resource allocation
- This forecasting system becomes a **strategic asset**, not just a model.

Even modest improvements in scheduling accuracy yield significant annual savings. The forecasting system becomes a strategic asset that enhances operational efficiency and financial performance.